BCD to Excess-3

Decimal	BCD				Excess-3			
	A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

Truth table for W

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

$$W = A + BD + BC$$

Truth table for X

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	X	X	X	X
10	0	1	X	X

$$X = B\bar{C}\bar{D} + \bar{B}D + \bar{B}C$$

BCD to Excess-3

Decimal	BCD				Excess-3			
	A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

Truth table for Y

AB \ CD	00	01	11	10
00	1	0	1	0
01	1	0	1	0
11	X	X	X	X
10	1	0	X	X

$$Y = CD + \bar{C}\bar{D}$$

Truth table for Z

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	X	X	X	X
10	1	0	X	X

$$Z = \bar{D}$$

8,4,-2,-1 to BCD

Decimal	BCD				8,4,-2,-1			
	W	X	Y	Z	A	B	C	D
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	1	1	1
2	0	0	1	0	0	1	1	0
3	0	0	1	1	0	1	0	1
4	0	1	0	0	0	1	0	0
5	0	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	0
7	0	1	1	1	1	0	0	1
8	1	0	0	0	1	0	0	0
9	1	0	0	1	1	1	1	1

Truth table for W

AB \ CD	00	01	11	10
00	0	X	X	X
01	0	0	0	0
11	X	X	1	X
10	1	0	0	0

$$W = AB + A\bar{C}\bar{D}$$

Truth table for X

AB \ CD	00	01	11	10
00	0	X	X	X
01	1	0	0	0
11	X	X	0	X
10	0	1	1	1

$$X = B\bar{C}\bar{D} + \bar{B}D + \bar{B}C$$

8,4,-2,-1 to BCD

Decimal	BCD				8,4,-2,-1			
	W	X	Y	Z	A	B	C	D
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	1	1	1
2	0	0	1	0	0	1	1	0
3	0	0	1	1	0	1	0	1
4	0	1	0	0	0	1	0	0
5	0	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	0
7	0	1	1	1	1	0	0	1
8	1	0	0	0	1	0	0	0
9	1	0	0	1	1	1	1	1

Truth table for Y

AB \ CD	00	01	11	10
00	0	X	X	X
01	0	1	0	1
11	X	X	0	X
10	0	1	0	1

$$Y = C\bar{D} + \bar{C}D$$

Truth table for Z

AB \ CD	00	01	11	10
00	0	X	X	X
01	0	1	1	0
11	X	X	1	X
10	0	1	1	0

$$Z = D$$

2,4,2,1 to 8,4,-2,-1

Decimal	2,4,2,1				8,4,-2,-1			
	A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	1	1	1
2	0	0	1	0	0	1	1	0
3	0	0	1	1	0	1	0	1
4	0	1	0	0	0	1	0	0
5	1	0	1	1	1	0	1	1
6	1	1	0	0	1	0	1	0
7	1	1	0	1	1	0	0	1
8	1	1	1	0	1	0	0	0
9	1	1	1	1	1	1	1	1

Truth table for W

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	X	X	X
11	1	1	1	1
10	X	X	1	X

$$W = A$$

Truth table for X

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	X	X	X
11	0	0	1	0
10	X	X	0	X

$$X = \bar{A}B + \bar{A}D + \bar{A}C + BCD$$

2,4,2,1 to 8,4,-2,-1

Decimal	2,4,2,1				8,4,-2,-1			
	A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	1	1	1
2	0	0	1	0	0	1	1	0
3	0	0	1	1	0	1	0	1
4	0	1	0	0	0	1	0	0
5	1	0	1	1	1	0	1	1
6	1	1	0	0	1	0	1	0
7	1	1	0	1	1	0	0	1
8	1	1	1	0	1	0	0	0
9	1	1	1	1	1	1	1	1

Truth table for Y

AB \ CD	00	01	11	10
00	0	1	0	1
01	0	X	X	X
11	1	0	1	0
10	X	X	1	X

$$Y = A\bar{C}\bar{D} + \bar{A}\bar{C}D + \bar{A}C\bar{D} + ACD$$

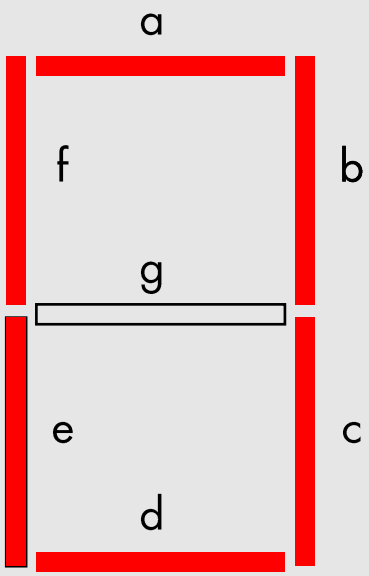
Truth table for Z

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	X	X	X
11	0	1	1	0
10	X	X	1	X

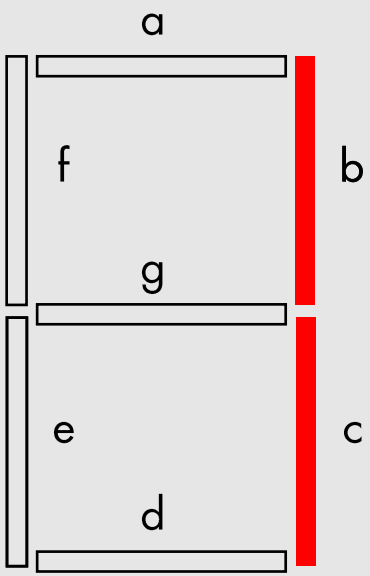
$$X = D$$

BCD to 7-segment

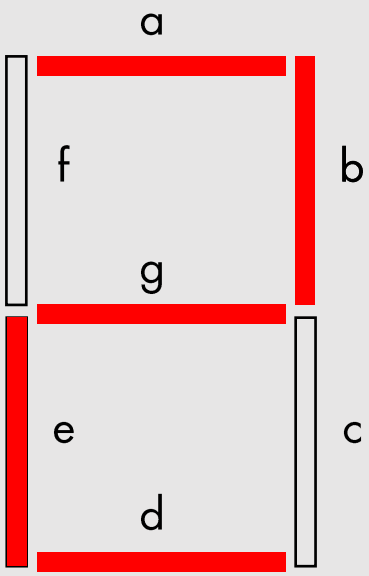
Decimal	BCD				7 segment						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1



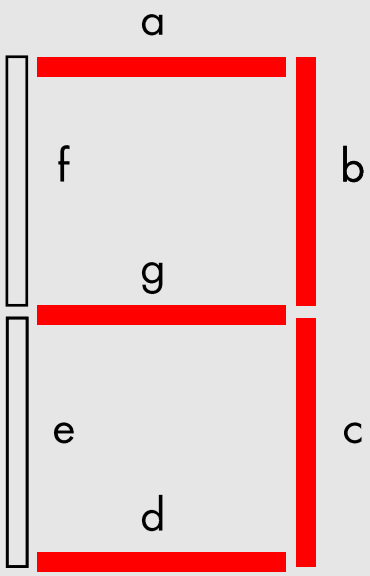
For 0



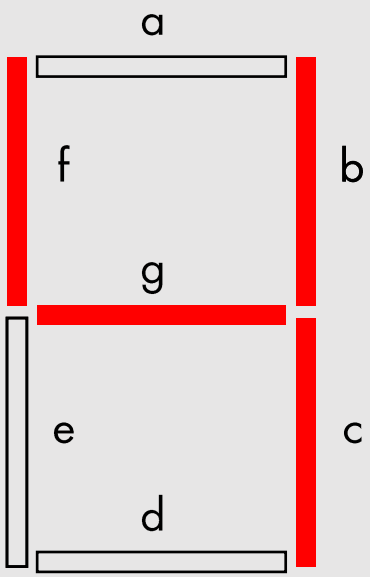
For 1



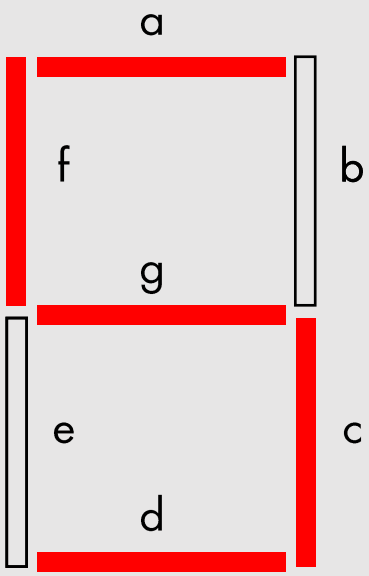
For 2



For 3



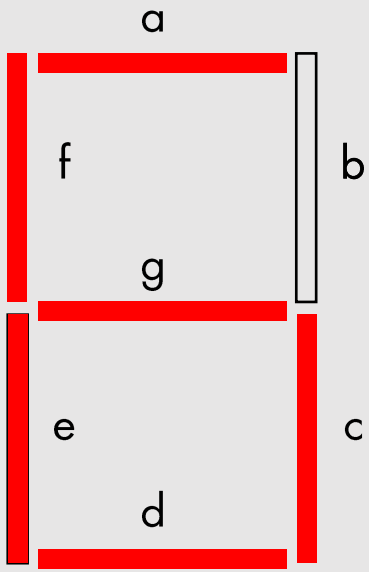
For 4



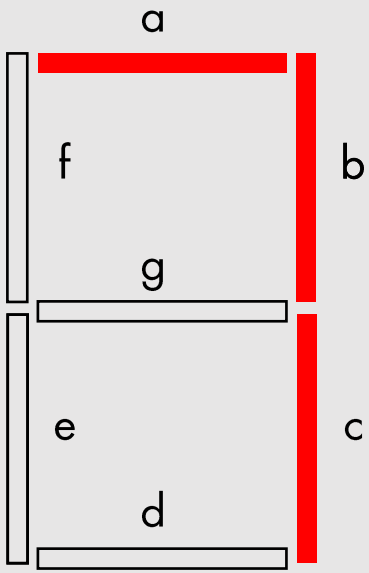
For 5

BCD to 7-segment

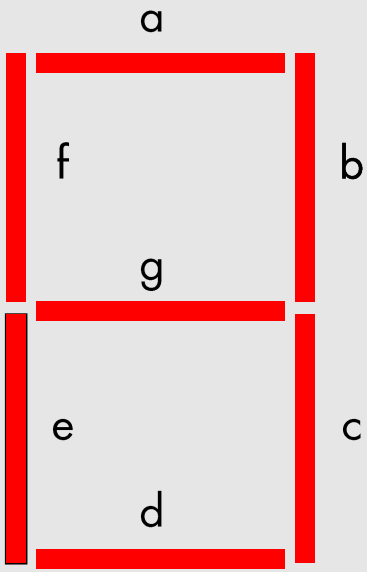
Decimal	BCD				7 segment						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1



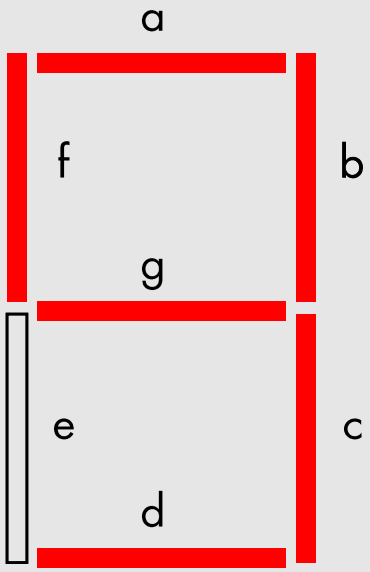
For 6



For 7



For 8



For 9

BCD to 7-segment

Decimal	BCD				7 segment						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

Truth table for a

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

$$a = A + BD + C + \bar{B}\bar{D}$$

Truth table for b

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	X	X	X	X
10	1	1	X	X

$$b = CD + \bar{C}\bar{D} + \bar{B}$$

Truth table for c

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	X	X	X	X
10	1	1	X	X

$$c = B + D + \bar{C}$$

Truth table for d

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	X	X	X	X
10	1	1	X	X

$$d = A + B\bar{C}D + C\bar{D} + \bar{B}\bar{D} + \bar{B}C$$

BCD to 7-segment

Decimal	BCD				7 segment						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

Truth table for e

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	X	X	X	X
10	1	0	X	X

$$e = C\bar{D} + \bar{B}\bar{D}$$

Truth table for f

AB \ CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

$$f = B\bar{D} + \bar{C}\bar{D} + A + B\bar{C}$$

Truth table for g

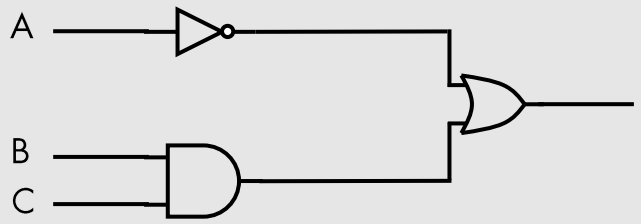
AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

$$g = A + B\bar{C} + C\bar{D} + \bar{B}C$$

NAND gate implementation

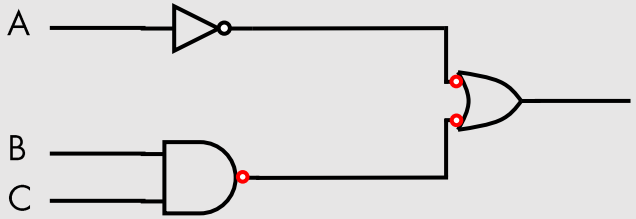
$Y = \overline{A} + BC$

STEP 1: Convert it into AOI (AND-OR-INVERTER) circuit



$Y = \overline{A} + BC$

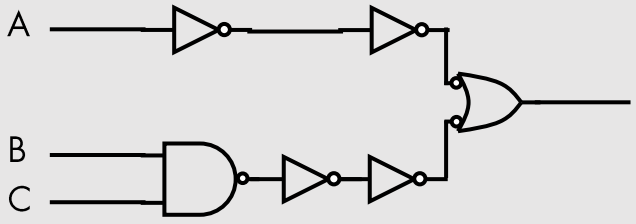
STEP 2: Put bubble on output of AND and input of OR



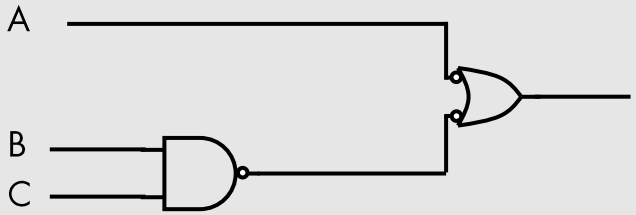
NAND gate implementation

$Y = \overline{A} + BC$

STEP 3: Add NOT gate for every bubble introduced



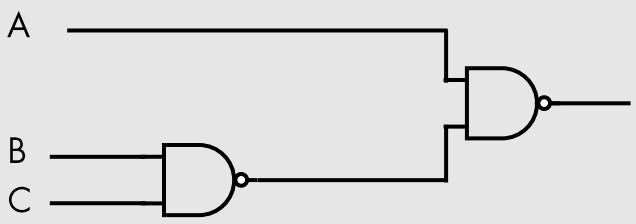
STEP 4: Remove NOT gates if there are two not gates one after another



NAND gate implementation

$Y = \overline{A} + BC$

STEP 5: Replace bubbled AND and bubbled OR gate

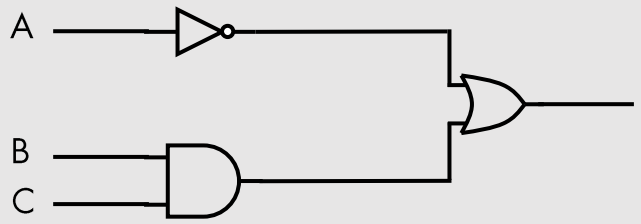


$Y = \overline{A} + BC$

NOR gate implementation

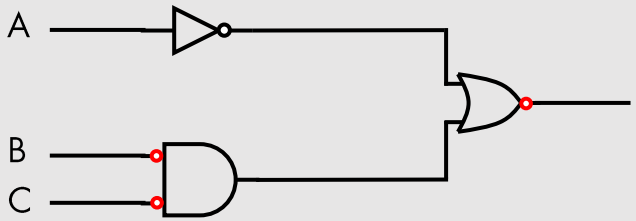
$Y = \overline{A} + BC$

STEP 1: Convert it into AOI (AND-OR-INVERTER) circuit



$Y = \overline{A} + BC$

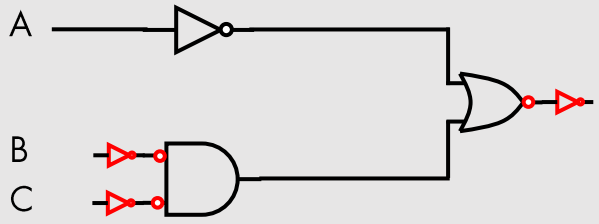
STEP 2: Put bubble on output of OR and input of AND



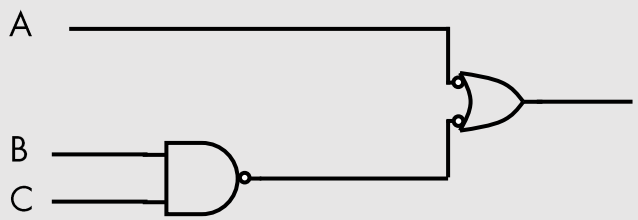
NAND gate implementation

$Y = \overline{A} + BC$

STEP 3: Add NOT gate for every bubble introduced



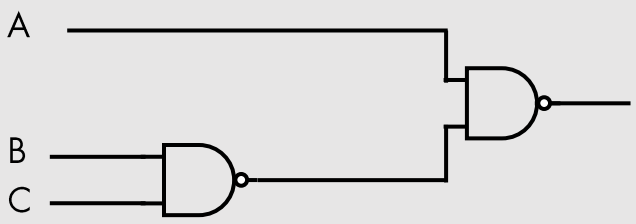
STEP 4: Remove NOT gates if there are two not gates one after another



NAND gate implementation

$Y = \overline{A} + BC$

STEP 5: Replace bubbled AND and bubbled OR gate



$Y = \overline{A} + BC$